

BOARD OF INTERMEDIATE EDUCATION, KARACHI

Computer Science Practical Journal

Class XI and Class XII

These are the laboratory activities named in the Sindh Curriculum for Computer Science, Grades XI-XII, 2019. Each practical states an aim, the steps, the program where the activity is code, what you should see, and two viva questions.

Programming activities use the same source as the lectures. Hardware activities are done with the teacher, with the mains lead removed before a case is opened.

The factor program, the pass-by-reference demonstration and the employee and string programs are printed in full in the Class XI lectures. Where one practical names several programs, the journal shows the first and the lecture shows the rest.

Practical XI-1

Parts of the system unit and the motherboard

Aim. Identify the processor, motherboard, memory, storage, power supply and ports, and say which block of the computer diagram each one belongs to.

You need

- A desktop system unit the teacher has opened, or a labelled photograph of a motherboard
- The block diagram from Unit 1

Steps

- 1 Unplug the computer before the case is opened. Do this only with the teacher.
- 2 Point to the power supply, the motherboard, the hard drive, the RAM and the processor fan.
- 3 On the motherboard, find the processor socket, the RAM slots, the CMOS battery, a port cluster and an expansion slot.
- 4 Write one sentence for each part saying whether it is processing, storage, memory or connection.

What you should observe

A labelled list of at least eight parts, each attached to its function.

Viva questions

- Which part keeps the clock when the mains lead is pulled out?
- Why is the processor fitted with a heat sink?

Practical XI-2

Seating memory and connecting drives

Aim. Remove and reseal a RAM module and trace the data and power cables of a drive.

You need

- An unplugged system unit
- A RAM module compatible with the board

Steps

- 1 Shut down, unplug the mains, and press the power button once.
- 2 Release the RAM clips, remove the module, and note the notch.
- 3 Press the module back into the same slot until both clips close.
- 4 Trace the drive's data cable to the board and its power cable to the power supply. Do not force a connector.

What you should observe

The machine still starts, and you can say which notch direction you used.

Viva questions

- Why must the mains lead be out?
- What does the notch prevent?

Practical XI-3

Boot options in firmware setup

Aim. Open the BIOS or UEFI setup and find the boot order without leaving a changed setting.

You need

- A lab computer
- The key the teacher gives for setup, often Del, F2 or F10

Steps

- 1 Restart and press the setup key before the operating system loads.
- 2 Find the screen that lists boot devices.
- 3 Read the first device in the list and write it down.
- 4 Exit without saving.

What you should observe

The name of the first boot device, and confirmation that you discarded changes.

Viva questions

- What is the BIOS for?
- What is the difference between a cold boot and a warm boot?

Practical XI-4

Files, folders, paths and search

Aim. Create, copy, move, rename and delete files, and find a file by name.

You need

- Windows with File Explorer

Steps

- 1 Create a folder CS-XI on the desktop.
- 2 Create a text file inside it, then copy the file, rename the copy, and move the copy to a second folder.
- 3 Write the absolute path of the renamed file, including the drive letter.
- 4 Delete nothing until you have searched for the file by part of its name and opened it from the search result.

What you should observe

Two folders, a renamed copy, and the path written in your journal.

Viva questions

- How does a copy differ from a move?
- What is an absolute path?

Practical XI-5

Attributes and the control panel

Aim. Set file attributes and locate hardware and software settings, dxdiag and msconfig.

You need

- The file created in XI-4

Steps

- 1 Open the file properties and set Read-only. Try to save a change and record what happens. Clear Read-only.
- 2 Set Hidden, refresh the folder, then show hidden files and clear the attribute.
- 3 From the control panel, find the display or sound settings and the list of installed programs. Change nothing you cannot reverse.
- 4 Run dxdiag and record the operating system name and the memory size. Run msconfig, look at Services, and close without saving.

What you should observe

The four attribute names, plus the operating system and RAM reported by dxdiag.

Viva questions

- What does the archive attribute mean?
- Why should msconfig be closed without a change unless you intend one?

Practical XI-6

Calculator

Aim. Use arithmetic operators and guard against division by zero.

You need

- A C++ compiler

Steps

- 1 Type the program, compile it, and run it twice.
- 2 First use 12 and 3. Then use 12 and 0.

Program

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;
    cout << "Sum = " << a + b << endl;
    cout << "Difference = " << a - b << endl;
    cout << "Product = " << a * b << endl;
    if (b != 0)
        cout << "Quotient = " << a / b << endl;
    else
        cout << "Cannot divide by zero" << endl;
    return 0;
}
```

What you should observe

The first run prints 15, 9, 36 and 4. The second run refuses the division.

Viva questions

- What is the difference between / and %?
- Why is the divisor tested?

Practical XI-7

Mark sheet with selection

Aim. Choose a grade with if and else-if, and reject an impossible mark.

You need

- A C++ compiler

Steps

- 1 Run the program with 76, with 40 and with 110.

Program

```
#include <iostream>
using namespace std;

int main() {
    int marks;
    cout << "Enter marks out of 100: ";
    cin >> marks;
    cout << "Grade: ";
    if (marks < 0 || marks > 100)
        cout << "Invalid" << endl;
    else if (marks >= 80)
        cout << "A" << endl;
    else if (marks >= 70)
        cout << "B" << endl;
    else if (marks >= 60)
        cout << "C" << endl;
    else if (marks >= 50)
        cout << "D" << endl;
    else
        cout << "Fail" << endl;
    return 0;
}
```

What you should observe

Grades B, Fail and Invalid.

Viva questions

- Why does the order of the else-if tests matter?
- Which structure would you use for a menu of exact day numbers?

Practical XI-8

Prime or composite

Aim. Use a loop and a decision to classify a number.

You need

- A C++ compiler

Steps

- 1 Test 17, 18 and 1.

Program

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a positive integer: ";
    cin >> n;
    if (n <= 1) {
        cout << n << " is neither prime nor composite" << endl;
        return 0;
    }
    bool isPrime = true;
    for (int i = 2; i <= n / 2; i++) {
        if (n % i == 0) {
            isPrime = false;
            break;
        }
    }
    if (isPrime)
        cout << n << " is prime" << endl;
    else
        cout << n << " is composite" << endl;
    return 0;
}
```

What you should observe

17 is prime, 18 is composite, and 1 is neither.

Viva questions

- Why does the loop start at 2?
- What does break do here?

Practical XI-9

Digits and factors

Aim. Isolate digits with integer division and list factors with a loop.

You need

- A C++ compiler

Steps

- 1 Run the digit program below with 2024.
- 2 Run Program 5.7, Factors of a number, from the Class XI lecture, with 12.

Program

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a four-digit number: ";
    cin >> n;
    if (n < 1000 || n > 9999) {
        cout << "Enter a four-digit number" << endl;
        return 0;
    }
    cout << n / 1000 << endl;
    cout << (n / 100) % 10 << endl;
    cout << (n / 10) % 10 << endl;
    cout << n % 10 << endl;
    return 0;
}
```

What you should observe

The digits of 2024 print as 2, 0, 2 and 4 on separate lines. The factor program prints 1 2 3 4 6 12.

Viva questions

- What does $n \% 10$ give?
- Why does the factor loop run from 1 to n ?

Practical XI-10

Functions: stars and the larger number

Aim. Write a function, call it, and return a value.

You need

- A C++ compiler

Steps

- 1 Build the star function below and call it with size 4.
- 2 Build Program 5.11, the larger function, from the Class XI lecture and call it with 14 and 22.

Program

```
#include <iostream>
using namespace std;

void drawStars(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }
}

int main() {
    int size;
    cout << "Enter size: ";
    cin >> size;
    drawStars(size);
    return 0;
}
```

What you should observe

Four rows of stars, growing from one star to four. larger(14, 22) returns 22.

Viva questions

- Which names are formal parameters?
- Where is the value of larger sent back?

Practical XI-11

Sort, search and average

Aim. Traverse an array to sort it, to find a value, and to average a list.

You need

- A C++ compiler

Steps

- 1 Run the sort and search program and look up 4.
- 2 Run the average program with four numbers: 10 20 30 40.

Program

```
#include <iostream>
#include <cstring>
using namespace std;

int main() {
    int a[] = {5, 1, 4, 2};
    int n = 4;
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - 1 - i; j++) {
            if (a[j] > a[j + 1]) {
                int temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
    cout << "Sorted: ";
    for (int i = 0; i < n; i++)
        cout << a[i] << " ";
    cout << endl;

    char names[3][10] = {"Sara", "Ali", "Hina"};
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2 - i; j++) {
            if (strcmp(names[j], names[j + 1]) > 0) {
                char temp[10];
                strcpy(temp, names[j]);
                strcpy(names[j], names[j + 1]);
                strcpy(names[j + 1], temp);
            }
        }
    }
    cout << "Names: ";
    for (int i = 0; i < 3; i++)
        cout << names[i] << " ";
    cout << endl;

    int key;
    cout << "Search for: ";
    cin >> key;
    int index = -1;
    for (int i = 0; i < n; i++) {
        if (a[i] == key)
            index = i;
    }
    if (index >= 0)
        cout << "Found at index " << index << endl;
    else
        cout << "Not found" << endl;
    return 0;
}
```

```
}
```

What you should observe

The numbers print as 1 2 4 5. The names print as Ali Hina Sara. 4 is found at index 2. The average is 25.

Viva questions

- Why is the first index 0?
- What does the function report if the value is absent?

Practical XI-12

Matrices, strings and an employee record

Aim. Add and multiply 3 by 3 matrices, reverse a name, and store an employee in a structure.

You need

- A C++ compiler

Steps

- 1 Run the matrix program below and check the first row of the sum and of the product.
- 2 Run Program 6.6 from the Class XI lecture with the name Ayesha.
- 3 Run Program 6.7, the employee structure, with Ali, Teacher and 55000.

Program

```
#include <iostream>
using namespace std;

int main() {
    int a[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
    int b[3][3] = {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}};
    int sum[3][3], product[3][3];

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++)
            sum[i][j] = a[i][j] + b[i][j];
    }
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            product[i][j] = 0;
            for (int k = 0; k < 3; k++)
                product[i][j] += a[i][k] * b[k][j];
        }
    }
    cout << "Sum row0: " << sum[0][0] << " " << sum[0][1]
        << " " << sum[0][2] << endl;
    cout << "Product row0: " << product[0][0] << " "
        << product[0][1] << " " << product[0][2] << endl;
    cout << "Product row2: " << product[2][0] << " "
        << product[2][1] << " " << product[2][2] << endl;
    return 0;
}
```

What you should observe

Sum row 2 2 3. Product equals the first matrix because the second is an identity. Ayesha reverses to ahseyA. The employee fields print back unchanged.

Viva questions

- What does the null character do in a C string?
- How do you reach a structure member?

Practical XI-13

Straight and crossover cables

Aim. Tell a straight-through cable from a crossover cable and know where each is used.

You need

- Two short lengths of twisted-pair cable
- RJ-45 plugs and a crimp tool, used by the teacher
- A cable tester if the lab has one

Steps

- 1 A straight-through cable uses the same colour order at both ends. The T568B order is white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
- 2 A crossover cable uses T568B at one end and T568A at the other. T568A swaps the orange and green pairs: white-green, green, white-orange, blue, white-blue, orange, white-brown, brown.
- 3 Use straight-through from a computer to a switch. The older direct computer-to-computer link used a crossover. Many modern ports detect the cable and adapt.
- 4 If you crimp, the teacher checks the order before the plug is pressed. Test continuity afterwards.

What you should observe

A note of both colour orders and which cable joins a PC to a switch.

Viva questions

- Which cable joins a PC to a switch?
- Name one guided medium other than twisted pair.

Practical XI-14

Network devices, IP settings and a small LAN

Aim. Identify the devices, read an IP configuration, and describe a central server against a peer arrangement.

You need

- The lab network
- A computer on that network

Steps

- 1 Point to the NIC, the switch, the router or modem, and any wireless access point.
- 2 Open the IP settings. Record the IPv4 address, the subnet mask and the default gateway.
- 3 A typical lab host might be 192.168.1.10 with mask 255.255.255.0 and gateway 192.168.1.1. Use the numbers your screen shows, not this example, if they differ.
- 4 Decide whether shared files come from one server (centralised) or from several student computers (decentralised, peer sharing). Write which one the lab uses and one consequence.

What you should observe

A sketch of the lab links and the three IP numbers from the computer you used.

Viva questions

- What does the default gateway do?
- Give one advantage of keeping shared files on one server.

Practical XII-1

Class Student

Aim. Hide age and percentage in a class and reach them through an object.

You need

- A C++ compiler

Steps

- 1 Compile the class, enter 17 and 88, and confirm that main does not assign to the private members directly.

Program

```
#include <iostream>
using namespace std;

class Student {
private:
    int age;
    float percentage;
public:
    void setData(int a, float p) {
        age = a;
        percentage = p;
    }
    void showData() {
        cout << "Age: " << age << endl;
        cout << "Percentage: " << percentage << endl;
    }
};

int main() {
    Student s;
    int age;
    float percentage;
    cout << "Enter age and percentage: ";
    cin >> age >> percentage;
    s.setData(age, percentage);
    s.showData();
    return 0;
}
```

What you should observe

Age 17 and percentage 88 printed by showData.

Viva questions

- Why is the data private?
- What is the object in this program?

Practical XII-2

Constructors for Student and Time

Aim. Initialise objects with a default constructor and a parameterised constructor.

You need

- A C++ compiler

Steps

- 1 Run the Student constructor program.
- 2 Run the Time program with 2 15 40 and notice the second object created with no arguments.

Program

```
#include <iostream>
using namespace std;

class Time {
private:
    int hours;
    int minutes;
    int seconds;
public:
    Time() {
        hours = 0;
        minutes = 0;
        seconds = 0;
    }
    Time(int h, int m, int s) {
        hours = h;
        minutes = m;
        seconds = s;
    }
    void show() {
        cout << "Time = " << hours << ":" << minutes
              << ":" << seconds << endl;
    }
};

int main() {
    int h, m, s;
    cout << "Enter hours, minutes and seconds: ";
    cin >> h >> m >> s;
    Time t(h, m, s);
    t.show();
    Time empty;
    empty.show();
    return 0;
}
```

What you should observe

Time prints 2:15:40 and the default object prints 0:0:0. The Student constructor program prints age 17 and percentage 91.

Viva questions

- How can you tell a constructor from an ordinary function?
- What is constructor overloading?

Practical XII-3

Inheritance, overloading and overriding

Aim. Reuse a base class, overload a member function, and override a virtual function.

You need

- A C++ compiler

Steps

- 1 Run the inheritance program with Ayesha 17 4521.
- 2 Run the polymorphism program and read both area lines and the show line.

Program

```
#include <iostream>
#include <string>
using namespace std;

class Base {
protected:
    string name;
    int age;
public:
    void setBase(string n, int a) {
        name = n;
        age = a;
    }
    void showBase() {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }
};

class DriveClass : public Base {
private:
    int roll;
public:
    void setDrive(string n, int a, int r) {
        setBase(n, a);
        roll = r;
    }
    void showDrive() {
        showBase();
        cout << "Roll: " << roll << endl;
    }
};

int main() {
    DriveClass student;
    string name;
    int age, roll;
    cout << "Enter name, age and roll: ";
    cin >> name >> age >> roll;
    student.setDrive(name, age, roll);
    student.showDrive();
    return 0;
}
```

What you should observe

Name, age and roll come from the derived object. The other program prints square 16, rectangle 20 and Drive show.

Viva questions

- What does protected allow?
- Why is show marked virtual?

Practical XII-4

Pointers

Aim. Show an address and a value, then change two variables through pointer parameters.

You need

- A C++ compiler

Steps

- 1 Run the basic pointer program and record the value before and after the write through the pointer.
- 2 Run the doubling program with 4 and 7.

Program

```
#include <iostream>
using namespace std;

int main() {
    int marks = 25;
    int *p = &marks;
    cout << "Value = " << marks << endl;
    cout << "Address stored in p = " << p << endl;
    cout << "Through pointer = " << *p << endl;
    *p = 40;
    cout << "New value = " << marks << endl;
    return 0;
}
```

What you should observe

The value changes from 25 to 40. The doubling program prints 8 and 14. The numeric address may differ each run.

Viva questions

- What does & return?
- What does the star do in *p and in the declaration int *p?

Practical XII-5

Text and binary files

Aim. Store a student in a text file and in a binary file, then read both back.

You need

- A C++ compiler
- A folder you are allowed to write in

Steps

- 1 Run the text program and open student.txt in an editor.
- 2 Run the binary program and compare the editor view of student.bin with the printed fields.

Program

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    ofstream outFile("student.txt");
    outFile << "Ayesha" << endl;
    outFile << 88 << endl;
    outFile.close();

    ifstream inFile("student.txt");
    string name;
    int marks;
    inFile >> name >> marks;
    cout << "Read: " << name << endl;
    cout << "Read: " << marks << endl;
    inFile.close();
    return 0;
}
```

What you should observe

The text file shows Ayesha and 88 as characters. The binary program prints Ayesha, 17 and 88.5. The binary file is not a document to edit.

Viva questions

- What does ios::app do differently from ios::out?
- What is EOF for?

Practical XII-6

SQL for a student result

Aim. Create a table, change its rows, and write a query that selects and projects.

You need

- Microsoft Access or any SQL environment the lab provides

Steps

- 1 Create STUDENT with StudentID as the primary key, StudentName as NOT NULL, and Dept.
- 2 Insert Ayesha of Science and Bilal of Science.
- 3 Update one department, then delete a row you created for practice.
- 4 Add RESULT with StudentID, SubjectID and Marks, and a foreign key to STUDENT if the software makes that easy.
- 5 Write a query that shows only the names of Science students, and a join that shows a name with a mark.

Program

```
CREATE TABLE STUDENT (  
    StudentID VARCHAR(6) PRIMARY KEY,  
    StudentName VARCHAR(40) NOT NULL,  
    Dept VARCHAR(20)  
);  
  
INSERT INTO STUDENT (StudentID, StudentName, Dept)  
VALUES ('S1', 'Ayesha', 'Science');  
  
SELECT StudentName  
FROM STUDENT  
WHERE Dept = 'Science';
```

What you should observe

The Science query returns Ayesha. The table refuses a second row with StudentID S1.

Viva questions

- Which clause selects rows?
- Which part of SELECT projects columns?

Practical XII-7

A short multimedia sequence

Aim. Combine text, an image and sound on layers, use two keyframes, export the result, and be ready to extend it to two minutes.

You need

- A video editor such as OpenShot, Shotcut or the package installed in the lab
- One diagram from these lectures and a short recording of your voice

Steps

- 1 Plan four scenes on paper. For this practical, build the first 20 to 30 seconds.
- 2 Set the canvas, put the diagram on a lower layer and a title on a higher layer.
- 3 Set a keyframe at the start and another two seconds later that moves the title slightly.
- 4 Add the voice recording. Export an MP4. If the software allows, export a second format as well.
- 5 Show the file to the teacher. Extend the same project to about two minutes for the full curriculum piece, then share it by the method the teacher sets.

What you should observe

An MP4 that shows a title over a diagram and plays your voice.

Viva questions

- What is a layer for?
- What is a keyframe for?